A1 | The trade-off is real; the candidate is conditional

A1 Decision | A2 Sizing | A3 Data pedigree | A4 Accuracy | A5 Sensitivity | A6 Economics

Does reducing cell count serve the same declared job?

Same-end 4 / 3 / 2-cell variants; identical load, duration and starting conditions. The table separates nominal behaviour from the required sensitivity gate.

Cells	Peak cell A	Peak °C	Lowest V	Cell cost ₹	Cell mass g
4	2.64	24.55	3.69	**₹4,824**	**280 g**
3	3.16	25.37	3.63	**₹3,618**	**210 g**
2	4.37	27.47	3.53	**₹2,412**	**140 g**

Source: Final rerun summary, nominal rows

What the change buys, and what it gives up		
Measure	**4 → 2 result**	**Interpretation**
Cell cost reduction	50%	Half the listed cell cost; a total build saving has not been calculated.
Cell mass reduction	50%	Half the cells' mass; no enclosure/board mass claim.
Peak current change	1.73 A	More current per cell; compare with operating limits and mismatch cases.
Lowest voltage change	-0.16 V	Less margin above the 3.3 V mission floor.
Temperature change	2.92 °C	A larger simulated rise; predictive temperature accuracy is unestablished.

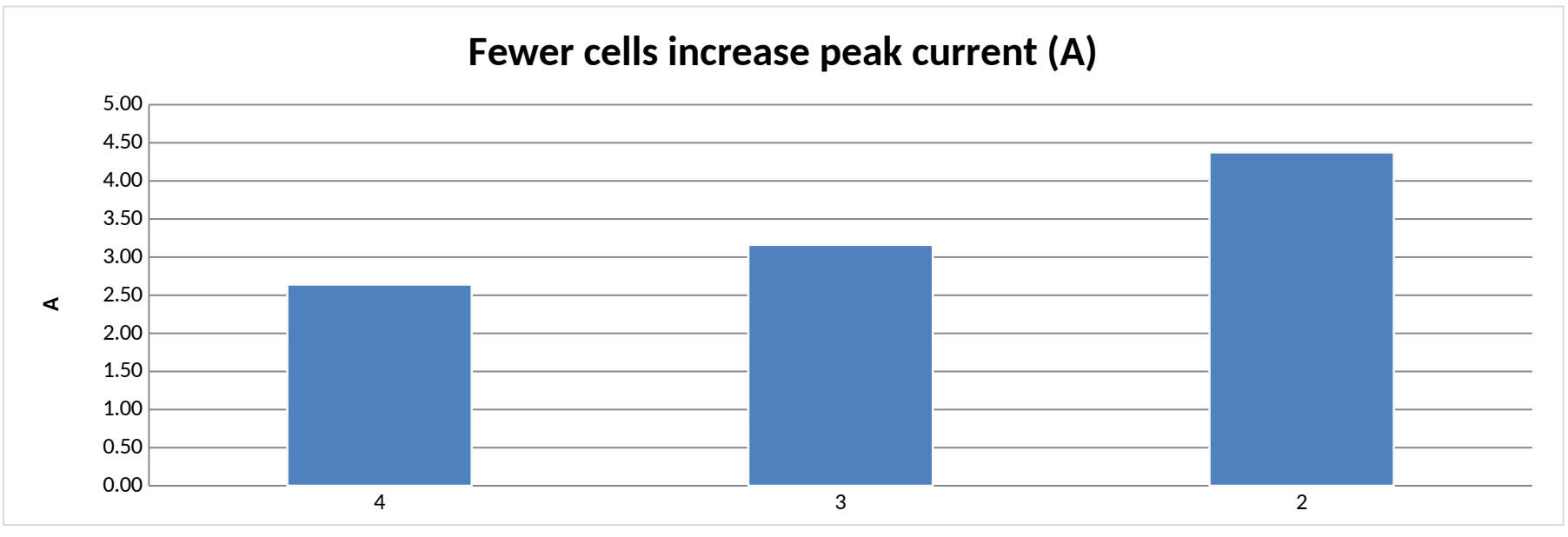
Decision: carry two cells forward as the smallest candidate tested. Keep three or four when redundancy or larger margins matter.

Method: coupled circuit solve → cell state update → irreversible/reversible heat → lumped temperature → updated cell parameters. Saved results retain the graph, settings and model/software identity.

Scope: a single parallel cell group. No ageing, mechanical packaging, core-temperature prediction, physical board protection or hardware acceptance is established.

Source: Cost and cell geometry

Source: Model equations and assumptions



A2 | Two is the smallest tested candidate, not a universal answer

How was the sizing decision derived?

Qualification requires every required Pi case to pass. Stress runs and random sensitivity samples are reported separately and are not probabilities of success.

1 | Convert the declared power scenario to the solver input

Power W	Assumed efficiency	Assumed floor V	Fixed-current proxy A
25	90%	3.30	8.418

25 W is the Pi 5 recommended supply envelope, not measured Pi draw. The solver holds 8.418 A constant; a true constant-power converter is not represented by this study.

Source: Pi 5 official supply specification

2 | Apply the frozen gate to all positions and required sensitivities

Cells	Required pass	Worst cell A	Lowest V	Worst °C	Gate result
1	0/15	Refused	No run	No run	Refused
2	33/33	5.49	3.50	31.78	Qualifies
3	42/42	3.86	3.62	27.69	Qualifies
4	51/51	3.34	3.66	26.12	Qualifies

Gate: complete 600 s; serve requested current; stay inside cell operating limits and model coverage; ≥3.3 V; calculate thermal state and close the first-law check. A closed energy balance detects accounting errors, not model bias.

3 | Compare the qualifying wiring choices, then state what could reverse the choice

Two-cell routing	Nominal peak A	Worst required A	12 A stress cases
baseline	4.37	5.49	30/31
thicker	4.31	5.44	30/31
opposite	4.21	5.33	31/31
combined	4.21	5.35	31/31

The study recommends opposite-end outputs. The slide holds same-end routing fixed to isolate cell-count trade-offs. Opposite-end routing requires physical return wiring; board layout may prevent it.

Two cells have no spare. Losing one leaves the refused one-cell scenario. Three remaining as two is a simulated contingency, not a tested fault/isolation system.

Source: Frozen protocol, tiers, mismatch magnitudes and ranking rule

Source: 4,006 final rerun cases and outcomes

A3 | A traceable parameter set still has transfer limits

Where do the cell and thermal numbers come from?

The prototype combines published cell tables, literature heat capacity, a fixture-specific cooling estimate and declared geometry. These have different evidential status.

Input	Value / derivation	Status	Consequence
Electrical response	PyECN LG M50T, 3 RC branches; 32 charge knots × 5 temperatures.	Published import	4–100% charge; 15–55 °C. Same cell type does not establish batch accuracy.
Heat capacity	58.5 J/K from Steinhardt et al., Table A.14.	Published reference	Held constant; source describes a different thermal observation/setup.
Heat loss to room	0.15249 W/K from test 53 rest steps 4 and 6.	Fixture calibration	A used four-cell rig. Transfer to a different cell assembly is assumed.
Cell size / mass	21.44 mm diameter × 70.8 mm height; maximum 70 g.	Draft manufacturer spec	Cell envelope only; no assembled pack layout, clearance or safety analysis.
Links / converter	Copper dimensions/contact resistance; 90% backup efficiency.	Declared assumptions	Not measured on the recommended build. Advertised board 95% is input step-down.
Low-charge regularization	Third-branch time constant held at 350.142 s below 18% charge.	Project modification	Unvalidated regularization, not an original measured value.

Source: Exact catalogue, parameter provenance and limitations

Source: PyECN pinned source commit 32540fe

Source: Heat capacity, DOI 10.1016/j.est.2021.103065

Dataset breadth and actual comparison coverage are different numbers		
Layer	**Count / use**	**What it establishes**
Published module dataset	54 experiments; 194 files checksum-verified	File identity/integrity; not 54 model validation tests.
Module comparison subset	2 recorded discharges: test 28 and test 53	Test 28 uses the same rig; test 53 also informed cooling. Neither is a blind independent holdout.
Adjacent-cell comparison	3 LG M50 cells × 2 currents = 6 discharges	Unrelated source and fixture; the M50 family is not confirmed identical to M50T.
Closed records	Reserved records remain closed	Not counted as validation. No outcomes decoded for this release.

So what: traceability lets a reviewer reconstruct the model. Independent predictive accuracy and transfer to a new board/fixture remain open questions.

Source: Module dataset: Piombo et al., CC BY 4.0

Source: Adjacent LG M50 dataset: Chen et al., CC BY 4.0

Source: Comparison allocation, independence and exclusions

A4 | Software checks pass; predictive accuracy remains unestablished

A1 Decision | A2 Sizing | A3 Data pedigree | A4 Accuracy | A5 Sensitivity | A6 Economics

What did the checks actually show?

Numerical consistency, repeatability and agreement with measurements answer different questions. A model can balance its energy perfectly and still predict a real cell poorly.

Check	Observed result	What follows
Independent numerical reference	15/15 synthetic electrical and heat cases against PyBaMM	Historical implementation comparison; no experimental temperature accuracy claim.
Frozen-study rerun	4,006/4,006 reproduced; 0 differing stored fields	27,832 numeric values unchanged on the final engine.
Energy accounting	3,964 calculated cases close; 42 cases refused	Internal accounting check; does not detect wrong material/model parameters.
Automated regression	515 Python pass, 8 known skips; 74 JavaScript pass	Software regressions covered; physical performance and provider reasoning need separate checks.

Published-data comparison: all 8 base discharges miss both predeclared bands

Discharge	V RMSE	V max error	Measured rise K	Model rise K	Voltage / temp
Module test 28	0.18	0.46	5.34	8.40	Miss / Miss
Module test 53	0.17	0.69	2.21	5.83	Miss / Miss
M50 cell02__2.5A	0.07	0.23	6.47	2.90	Miss / Miss
M50 cell02__5A	0.09	0.24	15.85	9.25	Miss / Miss
M50 cell03__2.5A	0.07	0.23	5.82	3.34	Miss / Miss
M50 cell03__5A	0.10	0.25	14.69	9.76	Miss / Miss
M50 cell04__2.5A	0.07	0.24	7.53	3.32	Miss / Miss
M50 cell04__5A	0.10	0.25	18.17	9.65	Miss / Miss

Bands fixed before comparison: voltage RMSE ≤0.05 V AND max error ≤0.15 V; thermal |peak-rise error| ≤ max(1 K, 25% of measured rise). Four module sensitivity reruns also miss both: 0/12 comparison cases inside both bands.

Current sharing: show the simple baseline beside the model

Record	Model mean error pp	Equal-share error pp	Busiest-cell match	Frozen gate
test28	1.13	2.38	62%	Miss
test53	0.21	0.21	48%	Miss

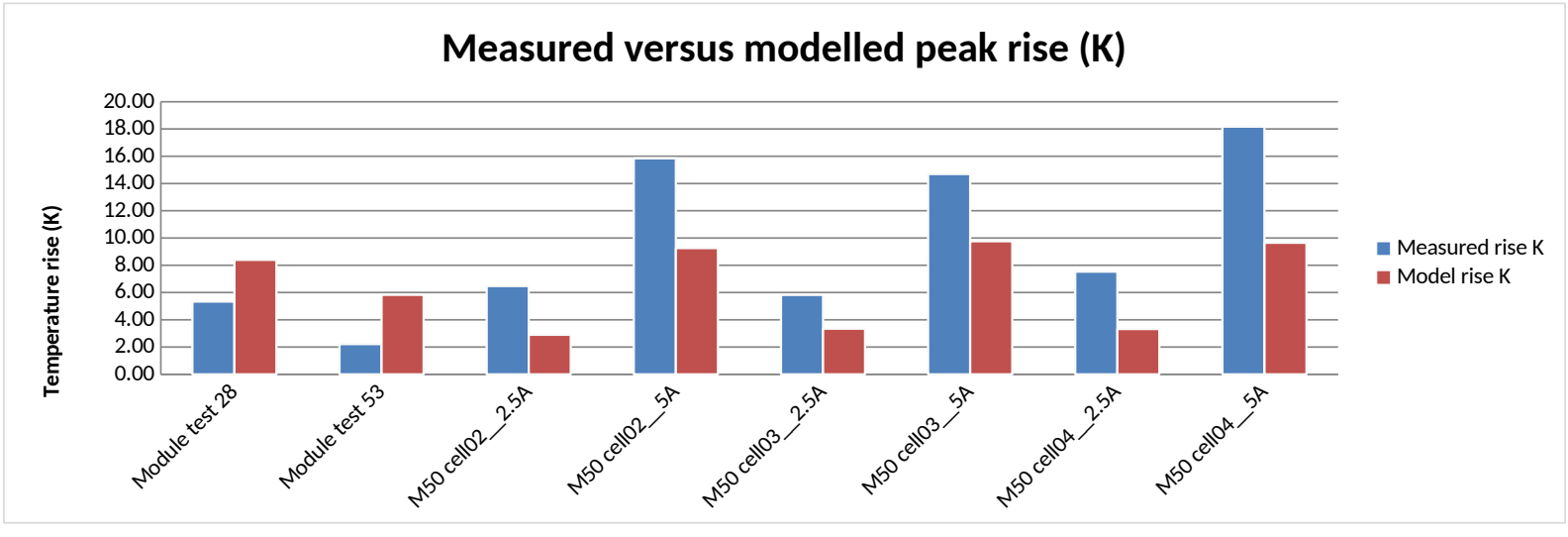
The selected early-discharge “34% versus 34%” slice is post hoc and is not a passed check. Sensor channel offsets and Hall-sum/cycler disagreement remain unresolved.

Source: Frozen measurement bands and method

Source: Every comparison result, including sensitivities

Source: Current sharing, equal-share baseline and post-hoc labels

Source: Final verification receipt



A5 | Sensitivity tests locate weak points; they are not confidence bands

Which assumptions move the candidate, and what remains untested?

Two-cell same-end Pi scenario. Change one declared input at a time. The frozen study also tests cell mismatch in every position and combinations separately.

Declared scenario	Peak cell A	Peak °C	Lowest V	Required result
nominal	4.37	27.47	3.53	Pass
cooling_half	4.37	29.11	3.53	Pass
thermal_pair_half	4.37	31.78	3.53	Pass
heat_capacity_low	4.37	27.53	3.53	Pass
contact_double	4.41	27.50	3.53	Pass
room_plus_2	4.37	29.02	3.53	Pass
all_charge_minus_2	4.37	27.29	3.51	Pass
efficiency 0.85	4.63	27.94	3.51	Pass
efficiency 0.95	4.14	27.07	3.55	Pass

Source: Scenario values are not probability distributions

Challenges outside the required gate		
Challenge	**Observed / unresolved**	**Decision consequence**
Higher current: 12 A stress	Same-end 2-cell: 30/31 deterministic stress cases; 59/62 combined samples. Opposite-end: 31/31 and 60/62.	Stress is report-only. Passing the Pi gate does not mean all loads or weak-cell combinations pass.
Lower real Pi demand	The 25 W scenario uses a recommended supply envelope, not a measured draw.	Measure actual demand later; a smaller load can change the minimum cell count.
Lost cell / open contact	One-cell scenario is refused. Three cells leaving two can serve the simulated scenario.	Two-cell choice has no spare; real fault currents, isolation and topology are unassessed.
Board integration and safety	Battery overcurrent, backup efficiency, charge sharing and sensing remain unverified. X1205 v1.1 cutoff is 2.4 V.	A candidate board does not implement the declared 3.3 V mission floor or prove cell protection.
Temperature forecast error	Published comparisons miss. Scaling exercises are scenarios, not statistical bounds.	No safety certification, confidence interval or guaranteed real maximum temperature.

Next evidence that changes the decision: actual Pi/board demand and efficiency; correct sensing and charge/discharge protection; independently measured voltage and temperature for the chosen arrangement. These are future physical checks, not a Round 1 claim.

Source: Detailed study and routing trade-offs

Source: Board screening, topology and operating-limit conditions

A6 | ₹5 lakh is a price hypothesis; revenue does not yet justify the build

What would the proposed pricing need to earn?

Blue numbers are editable assumptions. Black cells calculate. Student counts describe an audience; only paid labs and teams generate revenue. No customer has tested these prices.

Driver	Input	Meaning
Eligible specialised E&T colleges	2610	AISHE 2021–22 responding-college subset; not all engineering institutions.
Assumed adoption	5%	Planning scenario, not a demand forecast.
Lab price ₹ / year	50000	Proposed annual licence. Students free.
Team price ₹ / year	500000	Agreed planning price, up to 5 seats; not tested with customers.
Planning horizon years	5	Customers assumed present from year one in recovery illustration.

At 5% reach: labs **130** **Annual lab revenue ₹** **₹6,500,000**

Reconstruct the build estimate before discussing recovery (₹ crore)

Workstream	Person-years	Loaded people ₹ cr
Measured dataset and validation	12.50	2.44
Electrical and thermal solvers to pack scale	10	3.38
Electrochemistry, CFD and FEA	12	4.08
3D Design Studio and CAD exchange	17	3.98
Component library and versioned model graph	9	1.98
Simulation, results, failure search and comparison	12	2.84
AI copilot	3	0.66
Reports, projects, collaboration and security	10	2.04
Quality, documentation and support	9	1.26
Product, design and management	6	2.31
People subtotal		**24.96**
Lab capital; 3-year recurring; one-off	1.02	3.00 0.50
Contingency	20%	**5.90**
Full future-suite estimate ₹ cr		**35.38**
Stage 1: dataset + results + quality; full lab costs		**10.97**
Annual gross revenue to recover stage 1 over 5 years ₹ cr		**2.19**
Teams needed after assumed lab revenue (before running costs)		**31**

Annual revenue sensitivity (₹ crore): teams at ₹5 lakh plus labs at ₹50,000

Lab adoption	0 teams	10 teams	25 teams	50 teams
1%	0.13	0.63	1.38	2.63
3%	0.39	0.89	1.64	2.89
5%	0.65	1.15	1.90	3.15

Caution: gross recovery excludes post-build running, support, sales, tax, churn and ramp-up. The team market is unsized. Stage boundaries are planning allocations, not a funded execution plan. ₹35.38 cr is the future suite, not the existing prototype.

Cost basis: assumed staffing × proxy salary rates × 1.4 overhead, plus equipment/recurring costs and 20% contingency. Principal/physics rates are imperfect role proxies; effort, equipment and duration are unquoted estimates.

Source: All staffing quantities, proxy rate sources and cost assumptions

Source: AISHE audience counts, college denominator and adoption caveats

Source: Commercial assumptions and stage allocation rule